THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS TWO-CHAPTER TEST SET B

Chapters 9 & 10: **Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A figure that matches itself only after a full turn has an order of rotational symmetry of: 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 360

2. The Indian flag's Ashoka Chakra (24 spokes) has an order of rotational symmetry of: 1

(a) 6 **(b)** 24 **(c)** 8 **(d)** 12

3. Symmetry is commonly seen in: 1

(a) only prime numbers **(b)** only bar graphs

(c) rangoli and architecture **(d)** only fractions

4. The number of lines of symmetry of a regular pentagon is: 1

(a) 1 **(b)** 10 **(c)** 0 **(d)** 5

5. The angle of rotational symmetry of a regular octagon is: 1

(a) 90° **(b)** 45° **(c)** 60° **(d)** 30°

6. Which integer is neither positive nor negative? 1

(a) 1 **(b)** 10 **(c)** 0 **(d)** −1

7. If the temperature rises from −4°C by 7°C it becomes: 1

(a) −3°C **(b)** 11°C **(c)** 3°C **(d)** −11°C

8. The value of $8 + (-3)$ is: 1

(a) −5 **(b)** 3 **(c)** 5 **(d)** 7

9. The value of $-7 + (7)$ is: 1

(a) 2 **(b)** −2 **(c)** 0 **(d)** 1

10. The value of $-11 + (6)$ is: 1

(a) −5 **(b)** 5 **(c)** −7 **(d)** −3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the property of a reflection about the mirror line. 2

12. Name a shape with infinitely many lines of symmetry and say why. 2

13. What is the absolute value of −15 and of 15? 2

14. Find the distance between −7 and 3 on the number line. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How can you make a symmetrical design by folding and cutting paper? 3

16. A submarine at −200 m rises 75 m, then dives 40 m. Where is it? 3

17. List the quadrilaterals with 2 lines of symmetry. 3

OR

A kite has how many lines of symmetry? Explain. 3

18. Show on a number line: $-2 + 5$. 3

OR

Explain with an example why $(-a) + a = 0$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Complete the figure given its left half and a vertical mirror line. 5
(ii) State the symmetry.
(iii) Name a real object like it.

OR

(i) State three regular polygons with their orders of rotational symmetry. 5
(ii) Find the sum of angles of rotation for one full turn in a pentagon.

20. (i) Show the addition (−4) + 7 and (−4) − 7 on the number line. 5
(ii) Compare the results.

OR

(i) Explain why subtraction of an integer is adding its opposite. 5
(ii) Evaluate 5 − (−3) and (−5) − 3.

END OF QUESTION PAPER — Total: 40 Marks

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